SLA-Adaptive Autoscaling for Cloud Applications Using Kubernetes

1. Problem Statement

Cloud applications experience fluctuating workloads, making it difficult to allocate computing resources efficiently. Fixed resource allocation can lead to underutilization during low demand and degraded performance during peak demand. Conventional autoscaling mechanisms, such as Kubernetes Horizontal Pod Autoscaler (HPA), primarily rely on predefined resource utilization thresholds, which may not adequately adapt to changing workloads or service-level requirements. This project addresses the challenge of dynamically adjusting autoscaling thresholds to maintain application performance while optimizing resource utilization in a private cloud environment.

2. Objective of the Project

To implement and evaluate an SLA-aware autoscaling mechanism based on the research paper *SLA-Adaptive Threshold Adjustment for a Kubernetes Horizontal Pod Autoscaler*. The project aims to compare conventional threshold-based autoscaling with adaptive threshold adjustment, evaluating their impact on application performance, resource utilization, and service-level objective (SLO) violations.

3. Cloud Services Used

- **Proxmox VE:** Provides the private cloud infrastructure and virtual machines.
- **Ubuntu Server:** Hosts the Kubernetes cluster and application workloads.
- **K3s (Kubernetes):** Provides container orchestration and autoscaling through HPA.
- **Prometheus and Grafana:** Collect application and infrastructure metrics and visualize performance.
- **Python:** Implements the adaptive threshold adjustment mechanism.
- **Locust:** Generates controlled workloads for experimentation.

4. Brief Description of the Proposed Solution

A lightweight Kubernetes cluster will be deployed on Ubuntu virtual machines hosted on the existing Proxmox server. A containerized application will be deployed and configured with Kubernetes HPA. An adaptive autoscaling controller will be implemented using the threshold-adjustment principles described in the selected research paper. The controller will use monitored resource utilization and application performance metrics to adjust scaling thresholds dynamically. The system will be evaluated under varying workloads and compared against

conventional HPA. The existing Minecraft server will also be incorporated as an additional workload to study resource utilization and performance under real-world gaming activity. Since Minecraft is stateful, its integration will be designed appropriately rather than assuming that additional replicas automatically share the same game world.

5. Expected Outcome

The project is expected to deliver a functional private cloud environment with adaptive autoscaling, demonstrating how dynamic threshold adjustment can improve resource utilization while maintaining application performance. Experimental results will compare conventional and adaptive autoscaling using metrics such as CPU and memory consumption, response time, scaling delay, and SLO violations. The Minecraft workload will provide an additional practical evaluation of resource management under gaming workloads. The final deliverables will include the implementation, monitoring dashboard, experimental results, and a report comparing the findings with the selected research paper.

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